

NOTES ON KAN EXTENSIONS

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1. INTRODUCTION

This expository note on Kan extensions is intended as a reference for readers with basic knowledge of category theory. The main reference is [Rie17].

2. PRELIMINARIES

2.1. Slice categories. Let C be a category. Recall that the *slice category over* an object c in C , denoted C/c , is defined by:

- Objects: pairs (x, f) where c is an object of C and $f : x \rightarrow c$ is a morphism in C .
- Morphisms: a morphism $(x, f) \rightarrow (x', f')$ in the slice category is a morphism $g : x \rightarrow x'$ in C such that $f' \circ g = f$.

Similarly, we have the *slice category under* an object of C , denoted c/C , where objects are pairs (x, f) with x an object of C and $f : c \rightarrow x$ a morphism in C .

2.2. Category of elements. Slice categories are a special case of a more general construction, which we now describe. Let $F : C \rightarrow \mathbf{Set}$ be a functor. The *category of elements* of F , denoted $\int F$, is the category with:

- Objects: pairs (c, x) where c is an object of C and x is an element of the set Fc .
- Morphisms: a morphism $(c, x) \rightarrow (c', x')$ in the category of elements $\int F$ is a morphism $f : c \rightarrow c'$ in C such that $Ff(x) = x'$.

One can regard F as defining a set of structures on each object of the category C , and the category $\int F$ is a category of "structured objects" formed from C by selecting a choice of structure for each object. Morphisms in $\int F$ preserve the structure.

Example 2.1. The category of elements $\int C(c, -)$ of the covariant representable functor defined by c is the slice category c/C under c . Similarly, the category of elements $\int C(-, c)$ of the contravariant representable functor is the slice category C/c over c .

One can recover c as the initial object of the slice category under c ; a category of elements having this property implies that it is a slice category. More precisely:

Lemma 2.2. *A functor $F : C \rightarrow \text{Set}$ is representable if and only if its category of elements has an initial object.*

Proof. The proof is a straightforward application of the Yoneda lemma. $\square$

Example 2.3. Let $K : C \rightarrow D$ be a functor and d an object of D . The *left fiber* and *right fiber* categories over d are defined as, respectively:

$$K \downarrow d = \int D(K-, d) \quad d \downarrow K = \int D(d, K-)$$

These are also known as *comma categories*. The functor K has a left adjoint if and only if each right fiber category has an initial object. Dually, the functor K has a right adjoint if and only if each left fiber category has a terminal object.

2.3. Limits and colimits: definitions. Given categories J and C , a *J-shaped diagram* in a category C is a functor $F : J \rightarrow C$. These form a category, denoted C^J , where morphisms are natural transformations. We have the constant diagram functor $\Delta : C \rightarrow C^J$ taking an object c to the constant diagram with value c .

Definition 2.4. A *cone* over a J -shaped diagram $F : J \rightarrow C$ with *apex* $c \in C$ is a natural transformation $\lambda : \Delta c \rightarrow F$ from the constant diagram valued at c to F . Denoting by $\text{Cone}(c, F)$ the set of such cones, we have:

$$\text{Cone}(c, F) := C^J(\Delta c, F)$$

Treating the object of c as a variable, we have the functor $\text{Cone}(-, F) : C^{\text{op}} \rightarrow \text{Set}$. The *category of cones* over F , denoted Cone/F , is defined as the category of elements of this functor:

$$\text{Cone}/F := \int \text{Cone}(-, F) = \int C^J(\Delta-, F)$$

A *limit* of F , if it exists, is defined via either of the following equivalent descriptions:

- A representation for the functor $\text{Cone}(-, F)$, so that:

$$C(-, \lim F) \cong \text{Cone}(-, F)$$

- A terminal object of Cone/F .

A limit of $F : J \rightarrow C$ comes with a cone $\lambda : \Delta(\lim F) \rightarrow F$. This gives structure morphisms $\lambda_j : \lim F \rightarrow Fj$ mapping out from the limit to the image of any objects $j \in J$:

$$\begin{array}{ccc} & \lim F & \\ \lambda_j \swarrow & & \searrow \lambda_k \\ Fj & \xrightarrow{Ff} & Fk \end{array}$$

Key examples of limits include the product, intersection, kernel, equalizer, pullback, and inverse limit. Before examining properties of limits, we introduce the dual notion, namely colimits.

Definition 2.5. A cone under a J -shaped diagram $F : J \rightarrow C$ with *nadir* $c \in C$ is a natural transformation $\lambda : F \rightarrow \Delta c$ from F to the constant diagram valued at c . Denoting by $\text{Cone}(F, c)$ the set of such cones, we have:

$$\text{Cone}(F, c) := C^J(F, \Delta c)$$

Treating the object of c as a variable, we have the functor $\text{Cone}(F, -) : C \rightarrow \text{Set}$. The category of cones under F , denoted F/Cone , is defined as the category of elements of this functor:

$$F/\text{Cone} := \int \text{Cone}(F, -) = \int C^J(F, \Delta -)$$

A colimit of F , if it exists, is defined via either of the following equivalent descriptions:

- A representation for the functor $\text{Cone}(-, F)$, so that:

$$C(\text{colim} F, -) \cong \text{Cone}(F, -)$$

- An initial object of F/Cone .

A colimit of $F : J \rightarrow C$ is equipped with a cone $\lambda : F \rightarrow \Delta \text{colim}_J F$. This gives structure morphisms $\lambda_j : Fj \rightarrow \text{colim} F$ mapping into the colimit from the image of any object $j \in J$:

$$\begin{array}{ccc} Fj & \xrightarrow{Ff} & Fk \\ & \searrow \lambda_j & \swarrow \lambda_k \\ & \text{colim } F & \end{array}$$

Key examples of colimits include the coproduct, union, cokernel, coequalizer, pushout, and direct limit.

2.4. Limits and colimits: properties. We now collect results about limits and colimits.

Definition 2.6. A category C is *complete* if it contains all limits of small diagrams, i.e. diagrams $F : J \rightarrow C$ where J is a small category. A functor $C \rightarrow D$ is *continuous* if it preserves limits of small diagrams. Dually, a category C is *cocomplete* if it contains all colimits of small diagrams. A functor $C \rightarrow D$ is *cocontinuous* if it preserves colimits of small diagrams.

Lemma 2.7. If a limit exists for all functors in C^J , then a choice of limit defines a right adjoint $\lim : C^J \rightarrow C$ to the constant diagram functor $\Delta : C \rightarrow C^J$. Dually, if a colimit exists for all functors in C^J , then a choice of colimit defines a left adjoint $\text{colim} : C^J \rightarrow C$ to the constant diagram functor $\Delta : C \rightarrow C^J$.

Proof. The result follows from straightforward considerations building on the comments about fiber categories given in Example 2.3. $\square$

Lemma 2.8. *If J has an initial object, then a limit of any diagram $F : J \rightarrow C$ exists: take $\lim F$ to be the value of F at the initial object of J . Dually, if J has a terminal object, then a colimit of any diagram $F : J \rightarrow C$ exists: take $\operatorname{colim} F$ to be the value of F at the terminal object of J .*

Moreover, the identity functor $C \rightarrow C$ admits a limit (resp. colimit) if and only if C has an initial object (resp. terminal object).

Proof. The proof is a straightforward application of definitions. $\square$

Lemma 2.9 (Right adjoints preserve limits). *Let $F : C \rightleftharpoons D : G$ be an adjunction. Let $H : J \rightarrow D$ be a diagram in D so that $GH : J \rightarrow C$ is a diagram in C . If the limits of these diagrams exist, then the right adjoint G sends one to the other:*

$$G(\lim H) \simeq \lim(GH)$$

Proof. We use the Yoneda lemma and the adjunction (F, G) :

$$\begin{aligned} C(c, G \lim H) &\simeq D(Fc, \lim H) = \operatorname{Cone}_D(Fc, H) \\ &= D^J(\Delta Fc, H) \simeq C^J(\Delta c, GH) \\ &= \operatorname{Cone}_C(c, GH) = C(c, \lim GH) \end{aligned}$$

The identification $D^J(\Delta Fc, H) \simeq C^J(\Delta c, GH)$ may not be obvious at first glance, but is easily verified using the adjunction (F, G) . $\square$

Dually, we have:

Lemma 2.10 (Left adjoints preserve colimits). *Let $F : C \rightleftharpoons D : G$ be an adjunction. Let $H : J \rightarrow C$ be a diagram in C so that $FH : J \rightarrow D$ is a diagram in D . If the colimits of these diagrams exist, then the left adjoint F sends one to the other:*

$$F(\operatorname{colim} H) \simeq \operatorname{colim}(FH)$$

Lemma 2.11. [Functoriality] *Suppose we have a commutative diagram:*

$$\begin{array}{ccc} J & \xrightarrow{G} & J' \\ & \searrow F & \swarrow F' \\ & C & \end{array}$$

Then, assuming the limits and colimits exist, there are canonical maps

$$\lim F' \rightarrow \lim F \quad \operatorname{colim} F \rightarrow \operatorname{colim} F'$$

Proof. We can pull back a cone over F' to one over F . Specifically, if $\lambda : \Delta c \rightarrow F'$ is a cone over F' with apex c , then $j \mapsto \lambda_{Gj}$ defines a cone over F with apex c . Since the limit represents cones, we apply the Yoneda lemma to obtain a map $\lim F' \rightarrow \lim F$. Similar remarks apply for colimits. $\square$

3. KAN EXTENSIONS

We arrive to the main section of this note, wherein we introduce Kan extensions. There is a natural directional split here between left and right Kan extensions reflecting the duality between limits and colimits, which in turn reflects the directional nature of morphisms in a category.

3.1. Definitions. Let $K : C \rightarrow D$ be a functor and E a category. We have a pullback functor:

$$K^* : E^D \rightarrow E^C$$

which takes a functor $G : D \rightarrow E$ to its precomposition with K , so that $K^*G = G \circ K$. On the other hand, given a functor $F : C \rightarrow E$, it is less straightforward to extend F to D , i.e., to find a functor $G : D \rightarrow E$ making the following diagram commute up to a natural transformation:

$$\begin{array}{ccc} C & \xrightarrow{K} & D \\ F \downarrow & \swarrow G & \\ E & & \end{array}$$

We have choices. The **left option** is:

- Express the commutativity relation as a natural transformation $F \rightarrow G \circ K = K^*G$, so that we consider the fiber category $F \downarrow K^*$.
- An initial object of the category $F \downarrow K^*$, if it exists, is called the *left Kan extension* of F under K , denoted $\text{Lan}_K F$. It comes equipped with a natural transformation $F \rightarrow K^*\text{Lan}_K F$.
- The initiality of the left Kan extension in $F \downarrow K^*$ provides a natural isomorphism.

$$E^D(\text{Lan}_K F, -) \rightarrow E^C(F, K^* -)$$

- When it exists, the left Kan extension of F is given by the formula:

$$(\text{Lan}_K F)(d) = \text{colim} \left(K \downarrow d \xrightarrow{\pi} C \xrightarrow{F} E \right)$$

where $\pi : K \downarrow d \rightarrow C$ is the canonical projection functor.

- If the left Kan extension exists for all $F : C \rightarrow E$, then it defines a left adjoint $\text{Lan}_K : E^C \rightarrow E^D$ to the pullback functor K^* .

Dually, the **right option** is:

- Express the commutativity relation as a natural transformation $G \circ K = K^*G \rightarrow F$, so that we consider the fiber category $K^* \downarrow F$.

- A terminal object of the category $K^* \downarrow F$, if it exists, is called the *right Kan extension* of F under K , denoted $\text{Ran}_K F$. It comes equipped with a natural transformation $K^* \text{Ran}_K F \rightarrow F$.
- The terminal nature of the right Kan extension in $K^* \downarrow F$ provides a natural isomorphism:

$$E^D(-, \text{Ran}_K F) \rightarrow E^C(K^* -, F)$$

- When it exists, the right Kan extension of F is given by the formula:

$$(\text{Ran}_K F)(d) = \lim \left(d \downarrow K \xrightarrow{\pi} \mathcal{C} \xrightarrow{F} E \right)$$

where $\pi : d \downarrow K \rightarrow \mathcal{C}$ is the canonical projection functor.

- If the right Kan extension exists for all $F : \mathcal{C} \rightarrow E$, then it defines a right adjoint $\text{Ran}_K : E^C \rightarrow E^D$ to the pullback functor K^* .

The adjunctions (when all Kan extensions exist), can be summarized in the following diagram:

$$\begin{array}{ccc} & \text{Lan}_K & \\ & \curvearrowright & \\ E^D & \xrightarrow{K^*} & E^C \\ & \curvearrowleft & \\ & \text{Ran}_K & \end{array}$$

Remark 3.1. Let $g : d \rightarrow d'$ be a morphism in D . Then we have functors $K \downarrow d \rightarrow K \downarrow d'$ and $K \downarrow d' \rightarrow K \downarrow d$, commuting with the projections to $\mathcal{C}$. By Lemma 2.11, taking colimits is covariant, while taking limits is contravariant, so we get a map in the correct direction in both cases.

4. EXAMPLES

4.1. First examples. If K is the identity functor, then K^* and Ran_K and Lan_K are all the identity functor. If $K : J \rightarrow 1$ is the functor to a point and $F : J \rightarrow \mathcal{C}$, then $\text{Lan}_K F$ and $\text{Ran}_K F$ compute the colimit and limit of F , respectively, if they exist.

4.2. Group actions. For a group G , a G -set is a functor $\bullet/G \rightarrow \text{Set}$ from the one-point category defined by G to Set ; we denote this category of functors as $\text{Set}(G)$. Let H be a subgroup of G . We have the inclusion of one-point categories $\bullet/H \rightarrow \bullet/G$. Pullback along this functor defines the restriction functor $\text{Res} : \text{Set}(G) \rightarrow \text{Set}(H)$. The left and right adjoints are, respectively:

$$\text{Ind}(A) = \text{Lan}_{\text{Res}}(A) = (G \times A)/H$$

$$\text{Coind}(A) = \text{Ran}_{\text{Res}}(A) = \text{Set}(H)(G, A)$$

To clarify: $(G \times A)/H$ is the quotient of $G \times A$ by the action of H given by $h \cdot (g, a) = (gh^{-1}, h \cdot a)$ for $g \in G, a \in A$, and $h \in H$; its action of G is given by $g' \cdot (g, a) = (g'g, a)$.

A morphism $f : G \rightarrow A$ in $\text{Set}(H)$ is a morphism of sets such that $f(hg) = h \cdot f(g)$ for any $g \in G$ and $h \in H$; the action of G on such morphisms is $(g' \cdot f)(g) = f(g(g')^{-1})$. We have:

$$\begin{array}{ccc} & \text{Ind} & \\ \text{Set}(G) & \xrightarrow{\text{Res}} & \text{Set}(H) \\ & \text{Coind} & \end{array}$$

In particular, suppose H is the trivial group. Then $\text{Set}(H) = \text{Set}$ and $\text{Ind}(A) = G \times A$ is the coproduct of G -many copies of A , while $\text{Coind}(A) = \text{Set}(G, A) = A^G$ is the product of G -many copies of A .

4.3. Power sets. Let Bool be the category with two objects 0 and 1 and a single non-identity morphism $0 \rightarrow 1$. Let X be a set, regarded as a small discrete category, and let $\mathcal{P}(X)$ be the power set, reggraded as a poset category under inclusion. There is an equivalence of categories:

$$\text{Bool}^X \rightarrow \mathcal{P}(X)$$

taking $F : X \rightarrow \text{Bool}$ to the set $A_F = \{x \in X \mid Fx = 1\}$. For functors F and G in Bool^X , there is a natural transformation $F \rightarrow G$ if and only if $A_F \subseteq A_G$. Limits in $\mathcal{P}(X)$ are intersections, while colimits are unions.

Now let $f : X \rightarrow Y$ be a map of sets, regarded as small discrete categories. There is a pullback functor $f^{-1} : \mathcal{P}(Y) \rightarrow \mathcal{P}(X)$ taking a subset $B \subseteq Y$ to $\{x \in X \mid f(x) \in B\}$. Its left and right adjoints are given by, respectively:

$$\begin{aligned} f_*(A) &= \text{Lan}_{f^{-1}}(A) = \{y \in Y \mid \exists x \in f^{-1}(y) : x \in A\} \\ f_!(A) &= \text{Ran}_{f^{-1}}(A) = \{y \in Y \mid \forall x \in f^{-1}(y) : x \in A\} \end{aligned}$$

In other words, $f_*(A)$ is the direct image of A consisting of all elements of the form $f(x)$ for $x \in A$. By contrast, $f_!(A)$ is the shriek image of A , consisting of all elements of y whose fibers are contained in A . We have the diagram:

$$(4.1) \quad \begin{array}{ccc} & f_* & \\ \mathcal{P}(Y) & \xrightarrow{f^{-1}} & \mathcal{P}(X) \\ & f_! & \end{array}$$

The adjunctions (f_*, f^{-1}) and $(f^{-1}, f_!)$ reflect the following properties:

$$\begin{aligned} f_*(A) \subseteq B &\Leftrightarrow A \subseteq f^{-1}(B) \\ f^{-1}(B) \subseteq A &\Leftrightarrow B \subseteq f_!(A) \end{aligned}$$

4.4. Dependent sum and product. We again regard a set X as a small discrete category. There is an equivalence of categories:

$$\mathbf{Set}^X \rightarrow \mathbf{Set}/X$$

Taking a functor $F : X \rightarrow \mathbf{Set}$ to the coproduct of the sets Fx , with coprojection map to X ; that is, $\sum_{x \in X} Fx \rightarrow X$. We write an object of $\mathbf{Set}/X$ as a coproduct $S = \sum_{x \in X} S_x$. Limits and colimits in $\mathbf{Set}/X$ are evaluated pointwise¹.

Let $f : X \rightarrow Y$ be a map of sets. There is a pullback functor $\Delta_f : \mathbf{Set}/Y \rightarrow \mathbf{Set}/X$ given on objects by:

$$T = \sum_{y \in Y} T_y \quad \mapsto \quad T \times_Y X = \sum_{x \in X} T_{f(x)}$$

The definition of the functor on morphisms is straightforward. Its left adjoint is the dependent sum functor, and its right adjoint is the dependent product functor; these are given as Kan extensions:

$$\begin{aligned} \Sigma_f = \mathbf{Lan}_{\Delta_f} : \quad \sum_{x \in X} S_x &\quad \mapsto \quad S = \sum_{y \in Y} \left(\sum_{x \in f^{-1}(y)} S_x \right) \\ \Pi_f = \mathbf{Ran}_{\Delta_f} : \quad \sum_{x \in X} S_x &\quad \mapsto \quad \sum_{y \in Y} \left(\prod_{x \in f^{-1}(y)} S_x \right) \end{aligned}$$

(Again, the definitions of these functors on morphisms are clear.) We have:

$$(4.2) \quad \begin{array}{ccc} & \Sigma_f & \\ \swarrow & \Delta_f & \searrow \\ Y/\mathbf{Set} & \xrightarrow{\quad} & X/\mathbf{Set} \\ \nwarrow & \Pi_f & \nearrow \end{array}$$

As an aside, we remark on the definition of polynomial functors. A *polynomial span* is a diagram in $\mathbf{Set}$ of the form $V \xleftarrow{h} A \xrightarrow{g} E \xrightarrow{t} W$, and it gives rise to a functor on slice categories given by:

$$\Sigma_t \circ \Pi_g \circ \Delta_h : \mathbf{Set}/V \rightarrow \mathbf{Set}/W$$

This is known as a *polynomial functor*, and has the explicit form:

$$\Sigma_t \circ \Pi_g \circ \Delta_h(S) = \sum_{w \in W} \sum_{e \in E_w} \prod_{a \in A_e} S_{h(a)} = \sum_{w \in W} \sum_{e \in E_w} \prod_{v \in V} S_v^{A_e \cap A_v}$$

where $A_e = g^{-1}(e)$, $A_v = h^{-1}(v)$, and $E_w = t^{-1}(w)$. One can take the definition of polynomial functors as those arising from polynomial spans. Alternatively, polynomial endofunctors on $\mathbf{Set}$ can be defined as the smallest class of functors containing the identity functor and constant functors, and closed under taking arbitrary coproducts and constant exponentials. See [Jac16, Section 2.2].

¹For every $x \in X$, there is a functor $\mathrm{ev}_x : \mathbf{Set}/X \rightarrow \mathbf{Set}$ which picks out the fiber over x . It has a left and right adjoint given by the skyscraper functor $\mathbf{Set} \rightarrow \mathbf{Set}/X$ sending a set to itself with map to X factoring through the inclusion $\{x\} \hookrightarrow X$. Thus, the functor ev_x commutes with limits and colimits.

4.5. Density and codensity. Given a functor $F : C \rightarrow D$, consider the left Kan extension of F along itself:

$$T = \text{Lan}_F F : D \rightarrow D$$

We claim this is a comonad. To see this, first note that T is initial in the category $F \downarrow F^*$, so comes equipped with a natural transformation $\alpha : F \rightarrow TF$ of functors $C \rightarrow D$. Now, the identity endofunctor 1_D on D naturally defines an object of $F \downarrow F^*$ via the identity natural transformation $F \rightarrow F^*1_D = F$. Similarly, the endofunctor T^2 defines an object of $F \downarrow F^*$ via the natural transformation $F \rightarrow TF \rightarrow T^2F$ given by $T\alpha \circ \alpha$. From the initiality of T we obtain natural transformations $\eta : 1_D \rightarrow T$ and $\mu : T \rightarrow T^2$. One shows that these satisfy the axioms of a comonad on D , known as the *density comonad*. If F admits a right adjoint, the comonad of the adjunction FG coincides with the density comonad T .

Dually, given any functor $G : D \rightarrow C$, the codensity monad is the right Kan extension of G along itself: $T = \text{Ran}_G G$. If G admits a left adjoint F , then $\text{Ran}_G G$ is equivalent to the monad GF from the adjunction.

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- [Rie17] Emily Riehl, *Category Theory in Context*, Courier Dover Publications, 2017.